Artificial Neural Networks

INTRODUCTION

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- Introduction
- Machine Learning
  - Deep Learning
- Supervised Learning
- Neural Networks

Introduction

WHAT ARE NEURAL NETWORK

Computational models inspir ng: explicit rules for human brain

Consist of interconnected nodes (neuron

Learn from data to solve complex tasks

world problems are too complex for rule-based solutions

Neural networks can learn patterns from data

Examples of Hard Problems

- Weather prediction
- Answering free-form questions
- People identification
- Customized recommendations

Machine Learning

Study of algorithms that learn from experience

Performance improves with more data

Contrasts with static, rule-based systems

Voice assistants requires learning from audio data, not hand-coded rules

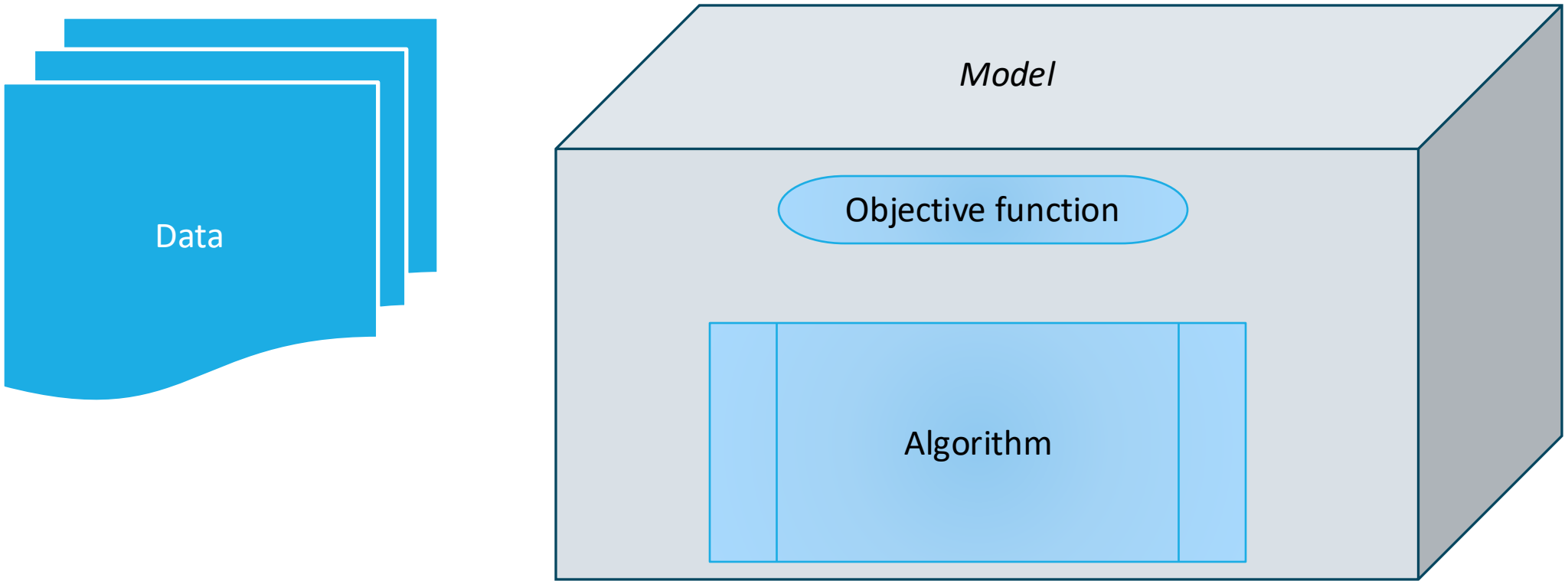
Programming with Data

- Instead of coding rules, provide examples (data)
- Model learns to map inputs to outputs

Deep Learning

- Focuses on models with many layers (deep architectures)
- Powers advances in vision, language, healthcare, and more

Machine Learning (components)



Machine Learning Data

Collection of examples (samples)

Each example has features (inputs) and often a label (target)

Data can be images, text, sensor readings, etc.

Data Representation

- Fixed-length vectors (e.g., image pixels)
- Variable-length data (e.g., text, reviews)
- Deep learning handles both types gracefully

Importance of Data quality

- More data enables more powerful models
- Poor data leads to poor models: "*Garbage in, garbage out*"
- Bias and underrepresentation can cause real-world harm

Machine Learning Models

Computational machinery for transforming data

Statistical models estimated from data

Deep learning: many layers of transformations (“deep”)

Objective functions

- Quantify how well the model is doing
- Lower is better (loss functions)

Algorithms

- Search for the best model parameters (optimization)
  - Update parameters to minimize loss (gradient descent)
- Iterative process



https://arxiv.org/html/2504.04242v1

Machine Learning
Deep Learning

A subset of machine learning concerned with models based on many-layered neural networks

Operations are learned jointly from data

Replacing the shallow models at the end of the traditional learning pipelines (and much of the domain-specific preprocessing)

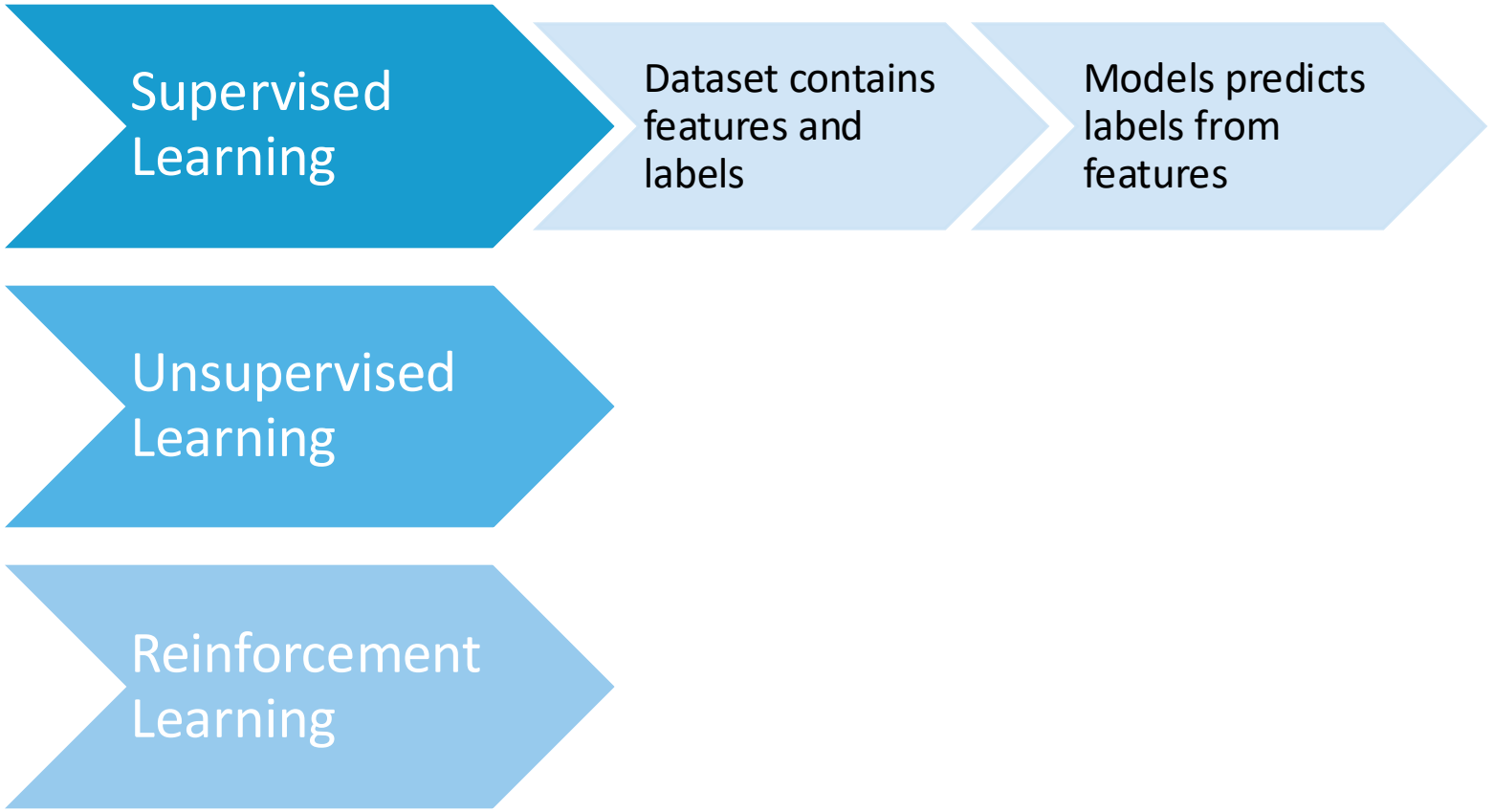
Parametric statistical descriptions to fully nonparametric models

- Scarce data requires reliance on simplified assumptions to obtain useful models
- Abundant data are better fitted with nonparametric models

Acceptance of suboptimal solutions

- Rapid progress in the development of practical algorithms
- Try things before proving them

Machine Learning problems



Supervised Learning

Regression
- Predicts continuous values

Classification
- Binary
- Multiclass
- Probabilistic

Multi-label classificati

Ranking
- Assign (relevance) scores to items
- Recommender systems

Sequence learning
- Inputs/outputs are sequences
- Context matters

What makes Neural Networks special?

Can approximate complex, non-linear functions
Learn hierarchical and complex representations
Adapt to new data and tasks



Lab Practice 0

Single Perceptron from scratch

Enrique Domínguez

DEEP LEARNING

Machine Learning Frameworks

Tensorflow

- Google
- Python
- Gran comunidad y popular
- Codificación extensa

PyTorch

- Facebook
- Arquitectura dinámica
- Python

CNTK

- Microsoft Cognitive Toolkit
- Python, C#, C++

MXNet

- Apache
- Python, C++, R, Julia, Javascript, Scala, Go
- Multi GPU

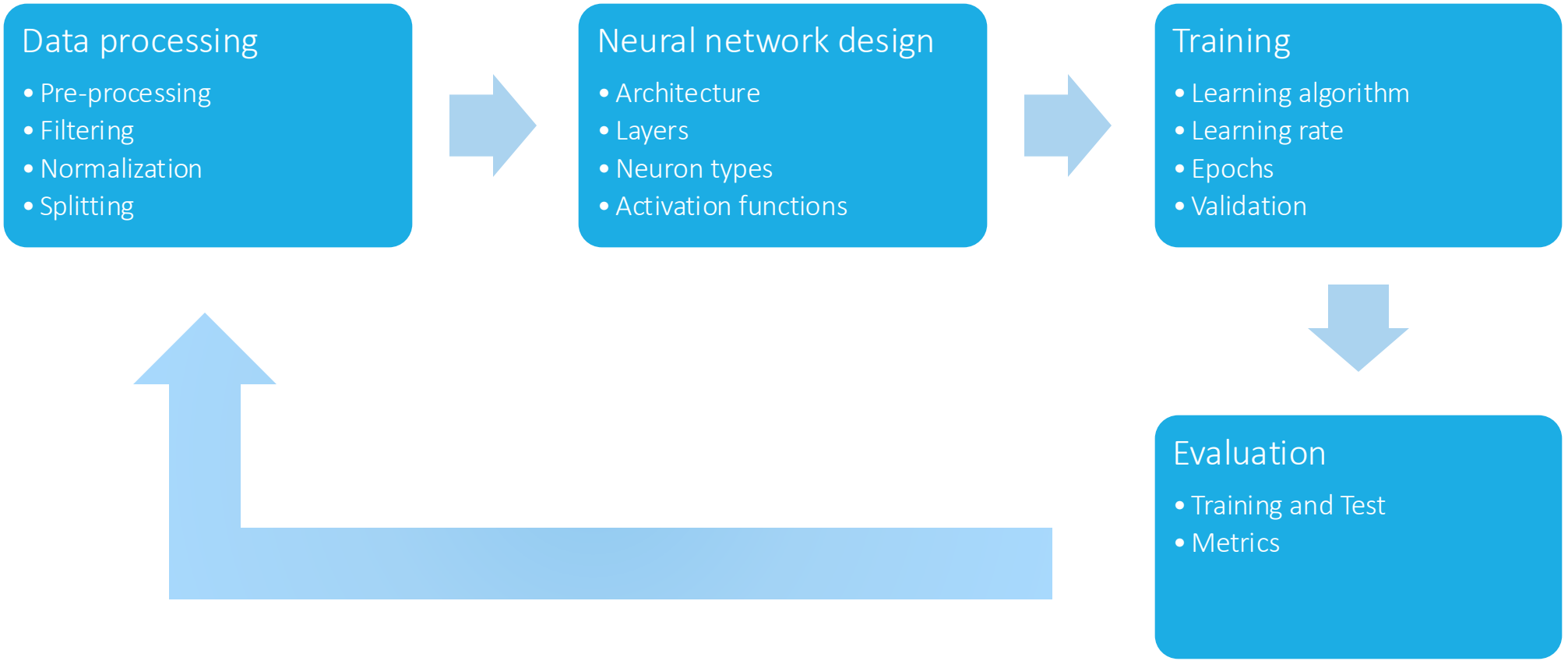
DeepLearning4J

- Java
- Hadoop y Spark

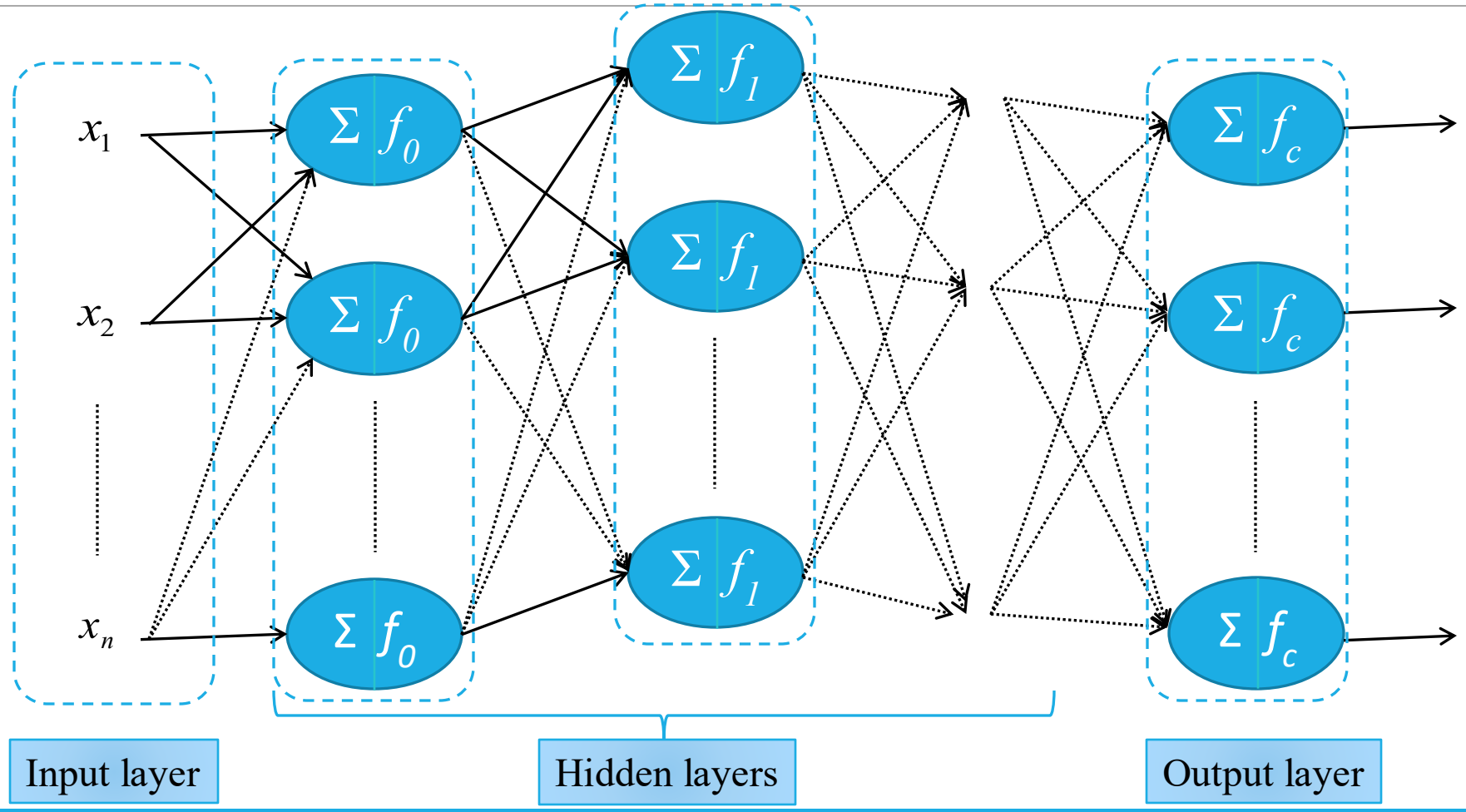
Keras

- API alto nivel
- Tensorflow, Theano, CNTK
- Python

Neural Networks Implementation steps



Neural Networks



Neural Networks Building

Architecture (topology)
- Sequential or Graph
- Layers, neurons

Activation functions
- Sigmoid, RELU, …

Data
- Inputs, Outputs
- Training, Validation, Test

Loss function (objective)
- Squared error, cross entropy, ...

Learning algorithm (optimizer)
- Gradient descent, …
- Parameters: Epochs, batch size, learning rate, ...

```
model = Sequential([
   Dense(10, input_dim=3, activation='relu'),
   Dense(1, activation='sigmoid')])

model.compile(optimizer='sgd',
       loss='mean_squared_error',
       metrics=['accuracy'])

model.fit(data, target,
       epochs=10,
       batch_size=16,
       validation_split=.1)

model.evaluate(testdata, testtarget)
```

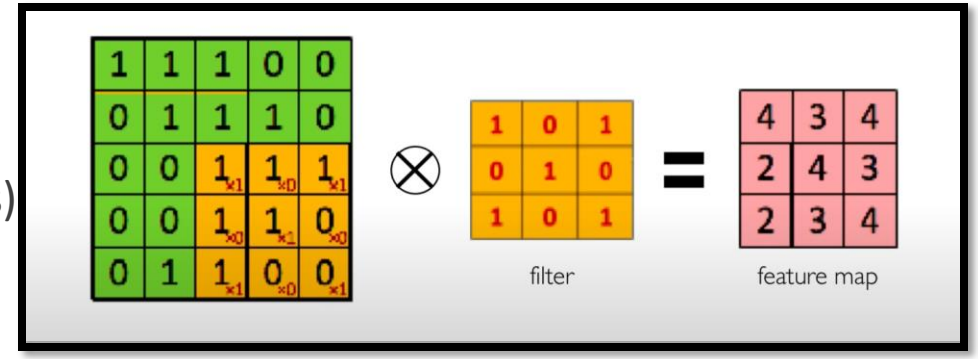
Neural Networks Layers

Linear (fully connected)

- Performs a weighted sum of the input
- With the activation function helps in learning complex patterns

Convolutional

- Applies convolution operations using filters
- Helps in capturing spatial features (e.g., edges, textures in images)



Recurrent

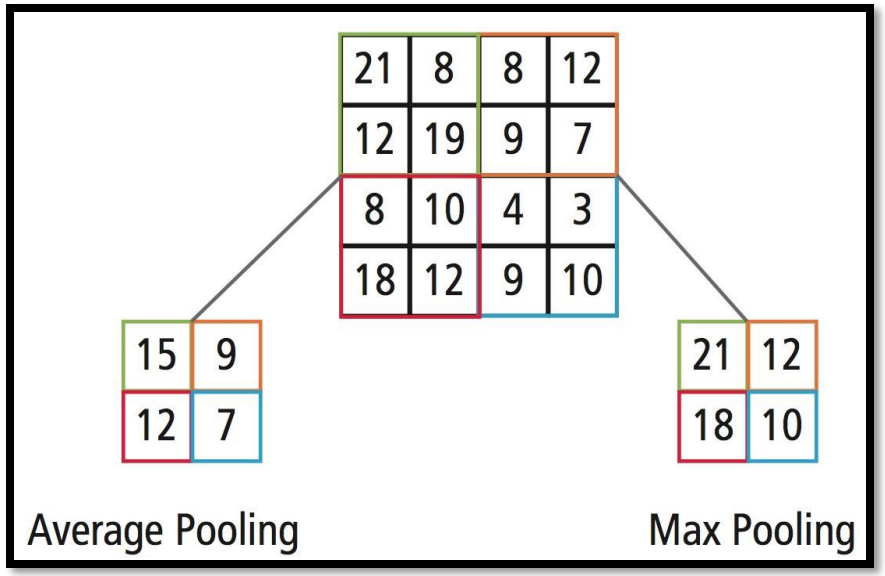
- Maintains state across time steps
- Helps in processing sequential data with temporal dependencies (e.g., time series, natural language)

Neural Networks Layers

Dropout
- Randomly drops a fraction of neurons during training
- Prevents overfitting

Pooling
- Reduces the spatial dimensions of the input

Padding
- Increases the spatial dimensions of the input



Normalization
- Normalizes the input (e.g., by subtracting the batch mean and dividing by the batch standard deviation)

Transformation
- Flatten, Reshape, ...

Neural Networks
Activation functions

Sigmoid functions

- Logistic
  - Smooth version of the step function
  - Infinitely differentiable
- Hyperbolic tangent function
  - Shifted and scaled version of logistic

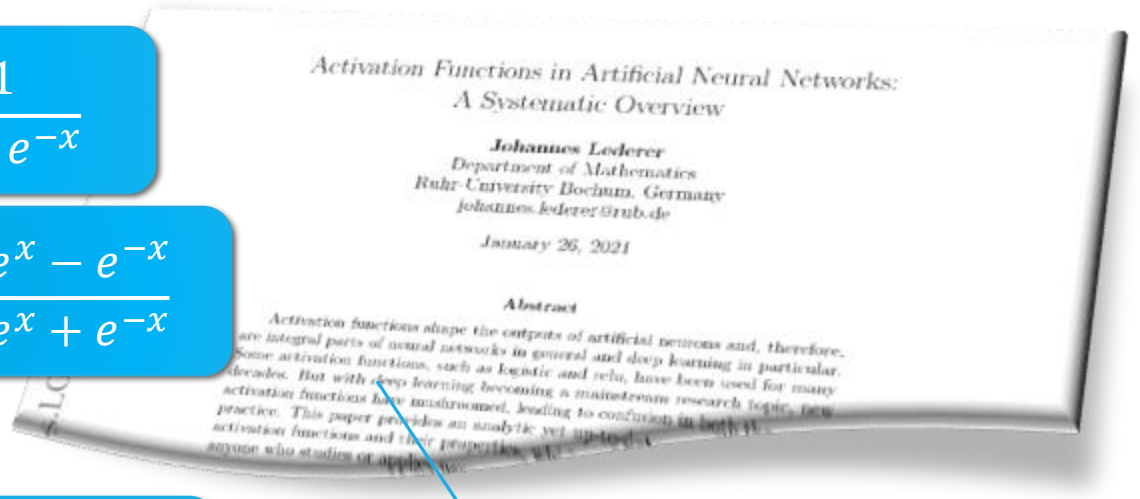
$$f(x) = \frac{1}{1 + e^{-x}}$$

$$f(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}}$$

Piecewise-Linear functions

- Linear
  - Identity function
- Ramp (relu)
  - Easy implementation
  - Not differentiable at 0 (vanishing gradient problem)

$$f(x) = \max(0,x)$$


Activation Functions in Artificial Neural Networks:
A Systematic Overview

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Abstract

https://arxiv.org/pdf/2101.09957

Neural Networks Learning Algorithms

Stochastic Gradient Descent

- Updates weights using the gradient from a small batch of data
- Faster updates, but with more variance

Adaptive Gradient Algorithm

- Adapts the learning rate for each parameter (weight)
- Smaller learning rates to frequently updated weights

Root Mean Square Propagation

- Adapts the learning rate by a moving average of squared gradients
- Prevents too small values of learning rate

Adaptive Moment Estimation

- Combines the momentum and RMSprop techniques
- More balanced and efficient for large datasets



Lab Practice 1

Performance study of different learning algorithms

Enrique Domínguez

DEEP LEARNING

Neural Networks
Training parameters

SETTINGS

Learning rate
- How much the parameters are adjusted in each iteration

Epochs
- Complete pass through the entire dataset

Batch size
- Number of samples used in each iteration

Validation set
- Subset of training set used to stop the training

LEARNABLE VARIABLES

Weights
- How much one neuron influences another
- Numerical values on the connections

Biases
- Shift the output of the activation function
- Value added to the weighted sum of inputs



Lab Practice 2

Adjusting the training parameters

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DEEP LEARNING